

ASSIGNMENT 1

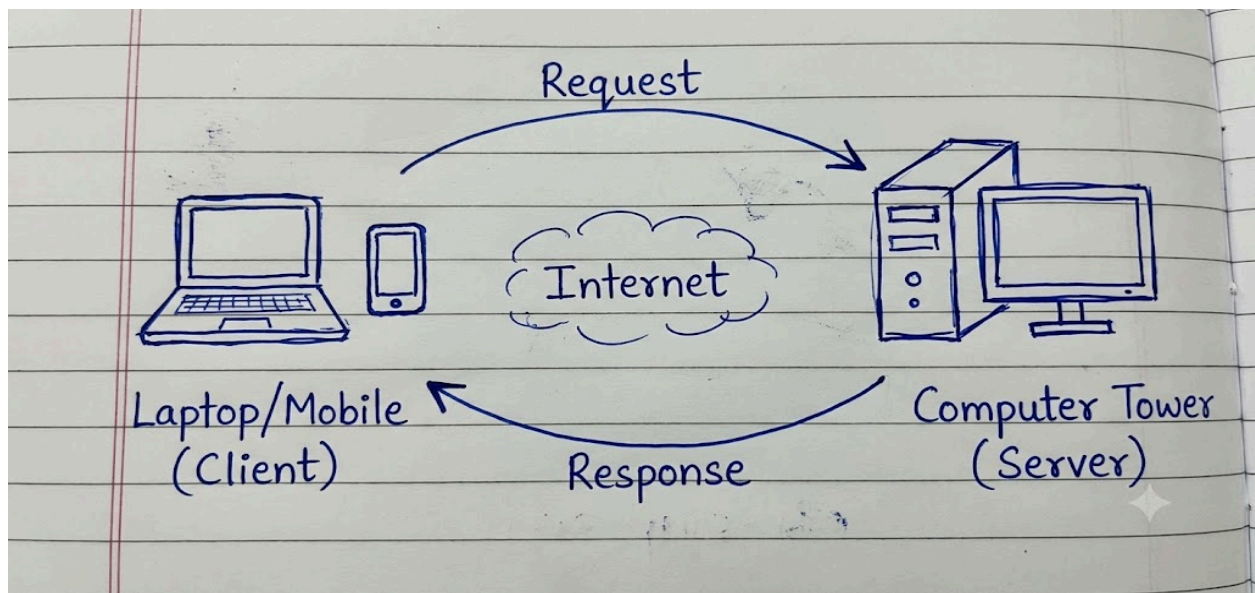
1. Explain the difference between frontend, backend, and full-stack development with suitable real-world examples.

Frontend Development: It focuses on the visual aspects of a website that users interact with (UI/UX). It is built using HTML, CSS, and JavaScript. Example: The layout, buttons, and product images you see on Amazon's website.

Backend Development: It involves the server-side logic, databases, and behind-the-scenes functionality that powers the website. Example: The system that verifies your login credentials or processes your payment when you buy a product on Amazon.

Full-Stack Development: This involves a combination of both frontend and backend development. A full-stack developer can build the user interface as well as manage the database and server logic. Example: A developer who designs the Amazon product page and also writes the code to save user orders in the database.

2. Create a simple diagram showing how the client-server model works in web architecture.



3. Describe how a browser requests and displays a web page from a web server.

1. URL Entry: The user types a website address (URL) into the browser.

2. DNS Lookup: The browser contacts a Domain Name System (DNS) server to find the IP address of the website's server.
3. HTTP Request: The browser sends an HTTP request to the web server asking for a specific web page.
4. Server Response: The server processes the request and sends back the requested files (HTML, CSS, JavaScript).
5. Rendering: The browser's rendering engine reads these files and paints the visual elements onto the screen for the user to view.

4. Identify and list the tools required to set up a web development environment. Explain the purpose of each.

1. Code Editor (e.g., VS Code): Used for writing, editing, and managing code files efficiently.
2. Web Browser (e.g., Google Chrome): Used for rendering the code to test and view how the website looks and functions.
3. Version Control System (e.g., Git & GitHub): Used to track changes in the code, maintain a history of modifications, and push the code to remote repositories for review and collaboration.

5. Explain what a web server is and give examples of commonly used servers.

A web server is a computer system or software application that stores website files (HTML documents, images, stylesheets) and delivers them to the user's browser over the internet upon request.

Commonly used examples: Apache HTTP Server, Nginx, and Microsoft Internet Information Services (IIS).

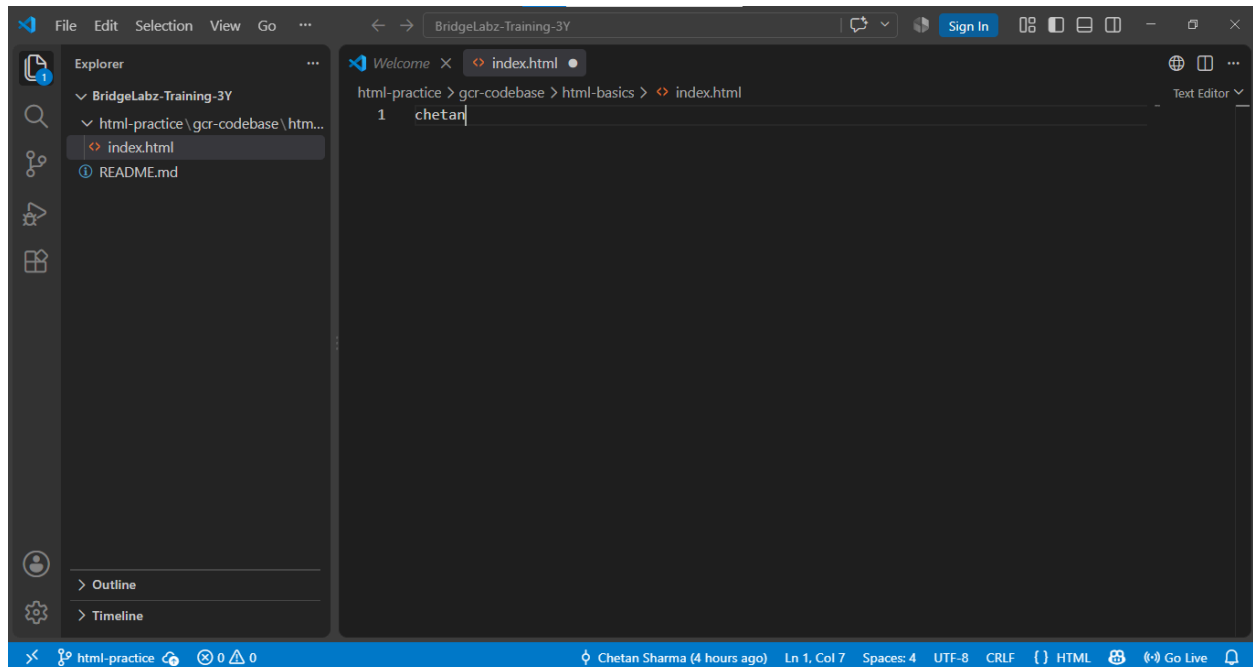
6. Define the roles of a frontend developer, backend developer, and database administrator in a project.

>Frontend Developer: Responsible for creating the user interface, ensuring the website is responsive (looks good on all devices), and optimizing the user experience.

>Backend Developer: Responsible for writing server-side logic, creating APIs, managing user authentication, and ensuring data security.

>Database Administrator (DBA): Responsible for designing the database architecture, ensuring data integrity, optimizing database performance, and managing data backups.

7. Install VS Code and configure it for HTML, CSS, and JavaScript development. Take a screenshot of the setup.



8. Explain the difference between static and dynamic websites. Provide an example of each.

* Static Website: The content is fixed and remains exactly the same for every user who visits the site. It is built using only HTML and CSS. Example: A company's basic informational portfolio website.

* Dynamic Website: The content changes dynamically based on user interaction, time, or data from a database. Example: Facebook or YouTube, where every user sees a customized feed and personalized

9. Research and list five web browsers. Explain how rendering engines differ between them.

A rendering engine is a software component that takes web content (HTML, XML, image files, etc.) and formats information to display it on the screen.

1. Google Chrome: Uses the 'Blink' engine.
2. Mozilla Firefox: Uses the 'Gecko' engine.
3. Apple Safari: Uses the 'WebKit' engine.

4. Microsoft Edge: Uses the 'Blink' engine.

5. Brave Browser: Uses the 'Blink' engine.

10. Draw a labeled diagram showing the basic web architecture flow — client, server, database, and APIs.

